Digital Logic Circuits (Spring 2026)

HW #2

11. Apply DeMorgan's theorems to the following:

(a) $\overline{\overline{(\overline{ABC})(\overline{EFG})} + \overline{(\overline{HIJ})(\overline{KLM})}}$ **(b)** $\overline{(A + \overline{B\overline{C}} + CD)} + \overline{\overline{BC}}$

(c) $\overline{\overline{\overline{(A + B)}\,\overline{(C + D)}\,\overline{(E + F)}\,\overline{(G + H)}}}$

11. (a)

$$\overline{\overline{\overline{(ABC)}}\,\overline{\overline{(EFG)}} + \overline{\overline{(HIJ)}}\,\overline{\overline{(KLM)}}} = \overline{\overline{\overline{ABC}} + \overline{\overline{EFG}} + \overline{\overline{HIJ}} + \overline{\overline{KLM}}}$$

$$= \overline{ABC + EFG + HIJ + KLM} = (\overline{ABC})(\overline{EFG})(\overline{HIJ})(\overline{KLM})$$

$$= (\overline{A} + \overline{B} + \overline{C})(\overline{E} + \overline{F} + \overline{G})(\overline{H} + \overline{I} + \overline{J})(\overline{K} + \overline{L} + \overline{M})$$

(b)

$$\overline{(A + \overline{B\overline{C}} + CD)} + \overline{\overline{BC}} = \overline{A}(\overline{\overline{B\overline{C}}})(\overline{CD}) + BC = \overline{A}(B\overline{C})(\overline{CD}) + BC$$

$$= \overline{A}B\overline{C}(\overline{C} + \overline{D}) + BC = \overline{A}B\overline{C} + \overline{A}B\overline{C}\,\overline{D} + BC = \overline{A}B\overline{C}(1 + \overline{D}) + BC$$

$$= \overline{A}B\overline{C} + BC$$

(c)

$$\overline{\overline{\overline{(A + B)}\,\overline{(C + D)}\,\overline{(E + F)}\,\overline{(G + H)}}}$$

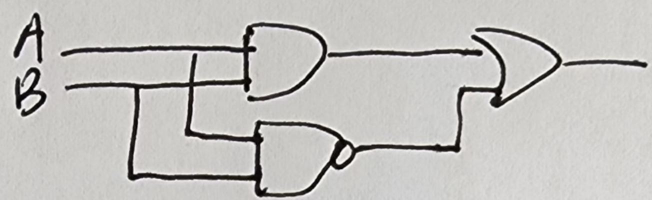
$$= (\overline{A + B})(\overline{C + D})(\overline{E + F})(\overline{G + H}) = \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}\,\overline{E}\,\overline{F}\,\overline{G}\,\overline{H}$$

15. Draw the logic circuit represented by each expression:

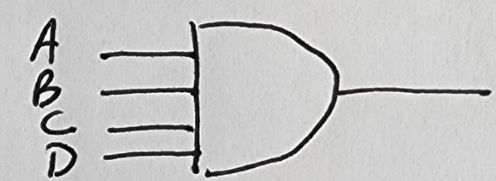
(a) $AB + \overline{AB}$ **(b)** $ABCD$

(c) $A + BC$ **(d)** $ABC + D$

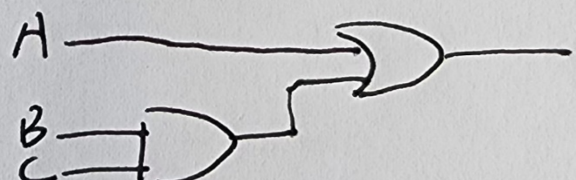
(a) $AB + \overline{AB}$



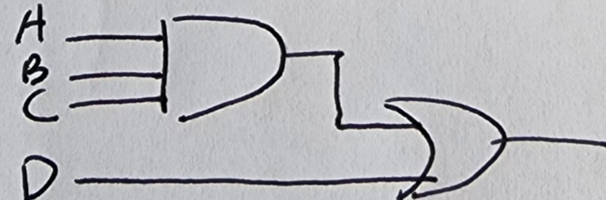
(b) $ABCD$



(c) $A + BC$



(d) $ABC + D$



21. Using Boolean algebra, simplify the following expressions:

(a) $CE + C(E + F) + \overline{E}(E + G)$ **(b)** $\overline{B}\,\overline{C}D + (\overline{B + C + D}) + \overline{B}\,\overline{C}\,\overline{D}E$

(c) $(C + CD)(C + \overline{C}D)(C + E)$ **(d)** $BCDE + BC(\overline{DE}) + (\overline{BC})DE$

(e) $BCD[BC + \overline{D}(CD + BD)]$

a) $CE + C(E + F) + \bar{E}(E + G) = CE + CE + CF + \bar{E}E + \bar{E}G = CE + CF += +\bar{E}G = \boldsymbol{CE + CF + \bar{E}G}$

b) $\bar{B}\bar{C}D + \overline{B + C + D} + \bar{B}\bar{C}\bar{D}E = \bar{B}\bar{C}D + \bar{B}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}E = \bar{B}\bar{C}D + \bar{B}\bar{C}\bar{D}(1 + E) = \bar{B}\bar{C}D + \bar{B}\bar{C}\bar{D} = \bar{B}\bar{C}(D + \bar{D}) = \boldsymbol{\bar{B}\bar{C}}$

c) $(C + CD)(C + \bar{C}D)(C + E) = C(1 + D)(C + D)(C + E) = C(C + D)(C + E) = (C + CD)(C + E) = C(1 + D)(C + E) = C(C + E) = C + CE = \boldsymbol{C}$

d) $BCDE + BC(\overline{DE}) + (\overline{BC})DE = BCDE + BC(\bar{D} + \bar{E}) + (\bar{B} + \bar{C})DE = BCDE + BC\bar{D} + BC\bar{E} + \bar{B}DE + \bar{C}DE = DE(BC + \bar{B} + \bar{C}) + BC\bar{D} + BC\bar{E} = DE(C + \bar{B} + \bar{C}) + +BC\bar{D} + BC\bar{E} = DE(1 + \bar{B}) + BC\bar{D} + BC\bar{E} = DE + +BC\bar{D} + BC\bar{E} = DE + BC(\bar{D} + \bar{E}) = DE + BC(\overline{DE}) = \boldsymbol{DE + BC}$

e) $BCD[BC + \bar{D}(CD + BD)] = BCBCD + BCD\bar{D}(CD + BD) = BCD + 0(CD + BD) = \boldsymbol{BCD}$

33. Develop a truth table for each of the SOP expressions:

(a) $\overline{A}B + AB\overline{C} + \overline{A}\,\overline{C} + A\overline{B}C$ **(b)** $\overline{X} + Y\overline{Z} + WZ + X\overline{Y}Z$

33. (a) $\overline{A}B + AB\overline{C} + \overline{A}\,\overline{C} + A\overline{B}C = \overline{A}BC + \overline{A}B\overline{C} + AB\overline{C} + \overline{A}\,\overline{B}\,\overline{C} + A\overline{B}C$

(b) $$\overline{X} + Y\overline{Z} + WZ + X\overline{Y}Z = \overline{W}\,\overline{X}\,\overline{Y}\,\overline{Z} + \overline{W}\,\overline{X}\,\overline{Y}Z + \overline{W}\,\overline{X}Y\overline{Z} + \overline{W}\,\overline{X}YZ$$
$$+ \overline{W}X\overline{Y}Z + \overline{W}XY\overline{Z} + W\overline{X}\,\overline{Y}\,\overline{Z} + W\overline{X}\,\overline{Y}Z$$
$$+ W\overline{X}Y\overline{Z} + W\overline{X}YZ + WX\overline{Y}Z + WXY\overline{Z} + WXYZ$$

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

W	X	Y	Z	Q
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

42. Expand each expression to a standard SOP form:

(a) $AB + A\overline{B}C + ABC$ **(b)** $A + BC$

(c) $A\overline{B}\,\overline{C}D + AC\overline{D} + B\overline{C}D + \overline{A}BC\overline{D}$ **(d)** $A\overline{B} + A\overline{B}\,\overline{C}D + CD + B\overline{C}D + ABCD$

43. Minimize each expression in Problem 42 with a Karnaugh map.

42. (a) $AB + A\overline{B}C + ABC = AB(C + \overline{C}) + A\overline{B}C + ABC$

$$= ABC + AB\overline{C} + A\overline{B}C + ABC$$

$$= ABC + A\overline{B}C + AB\overline{C}$$

(b) $A + BC = A(B + \overline{B})(C + \overline{C}) + (\overline{A} + A)BC = (AB + A\overline{B})(C + \overline{C}) + (\overline{A} + A)BC$

$$= ABC + AB\overline{C} + A\overline{B}C + A\overline{B}\,\overline{C} + \overline{A}BC + ABC$$

$$= ABC + AB\overline{C} + A\overline{B}C + A\overline{BC} + \overline{A}BC$$

(c) $A\overline{B}\,\overline{C}D + AC\overline{D} + B\overline{C}D + \overline{A}BC\overline{D}$

$$= A\overline{B}\,\overline{C}D + A(B + \overline{B})CD + (A + \overline{A})B\overline{C}D + \overline{A}BC\overline{D} =$$

$$= A\overline{B}\,\overline{C}D + ABCD + A\overline{B}CD + AB\overline{C}D + \overline{A}B\overline{C}D + \overline{A}BC\overline{D}$$

(d) $A\overline{B} + A\overline{B}\,\overline{C}D + CD + B\overline{C}D + ABCD$

$$= A\overline{B}(C + \overline{C})(D + \overline{D}) + A\overline{B}\,\overline{C}D + (A + \overline{A})(B + \overline{B})CD + (A + \overline{A})B\overline{C}D + ABCD$$

$$= A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}D + ABC\overline{D} + ABCD + A\overline{B}\,\overline{C}D + ABCD + A\overline{B}CD + \overline{A}BCD$$

$$+ \overline{A}\,\overline{B}CD + AB\overline{C}D + \overline{A}B\overline{C}D + ABCD$$

$$= A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}\,\overline{C}D + ABC\overline{D} + ABCD + A\overline{B}CD + \overline{A}BCD + \overline{A}\,\overline{B}CD + AB\overline{C}D + \overline{A}B\overline{C}D$$

$$= \overline{A}\,\overline{B}CD + \overline{A}B\overline{C}D + \overline{A}BCD + A\overline{B}C\overline{D} + A\overline{B}\,\overline{C}D + A\overline{B}CD + AB\overline{C}D + ABCD + ABC\overline{D}$$

43. See Figure 4-14.

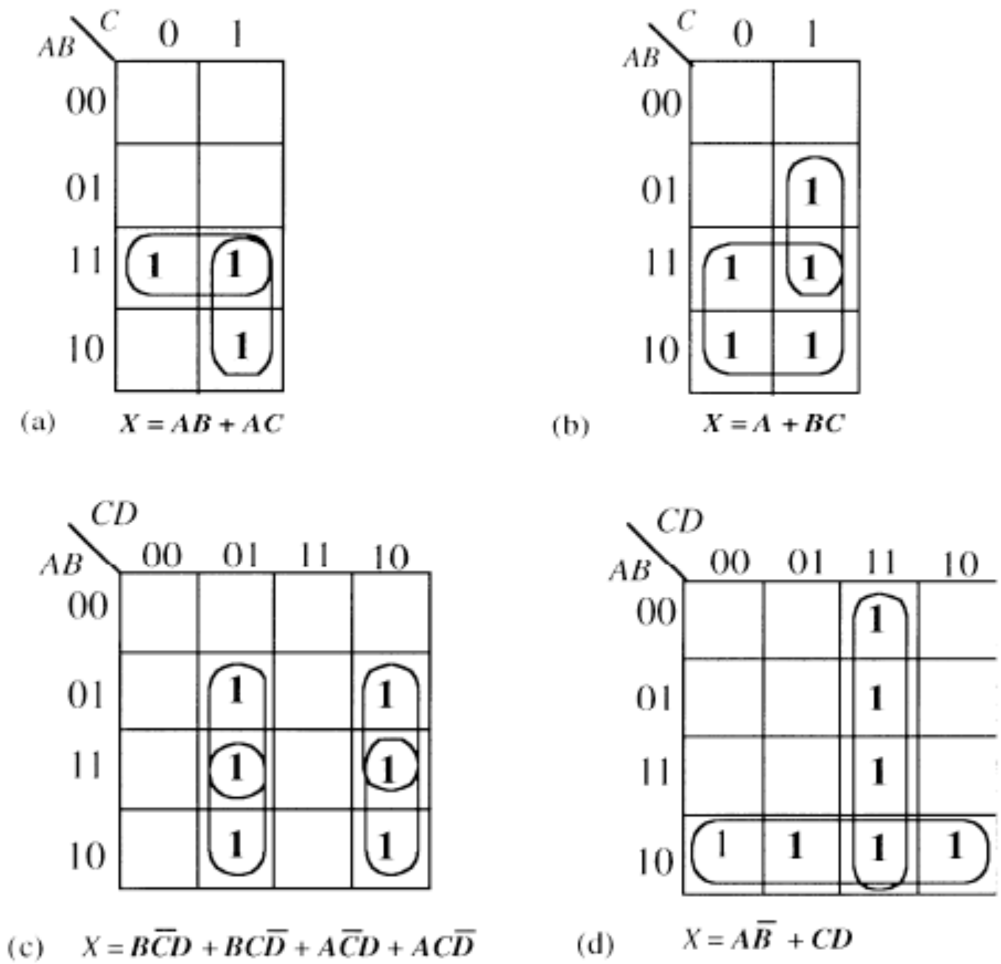


FIGURE 4-14

49. Use a Karnaugh map to simplify each expression to minimum POS form:

(a) $(A + \overline{B} + C + \overline{D})(\overline{A} + B + \overline{C} + D)(\overline{A} + \overline{B} + \overline{C} + \overline{D})$

(b) $(X + \overline{Y})(W + \overline{Z})(\overline{X} + \overline{Y} + \overline{Z})(W + X + Y + Z)$

49. (a) $(A+\overline{B}+C+\overline{D})(\overline{A}+B+\overline{C}+D)(A+\overline{B}+\overline{C}+\overline{D})$

0101 1010 1111

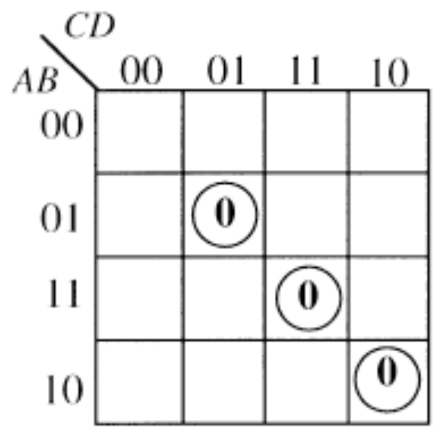


FIGURE 4-22

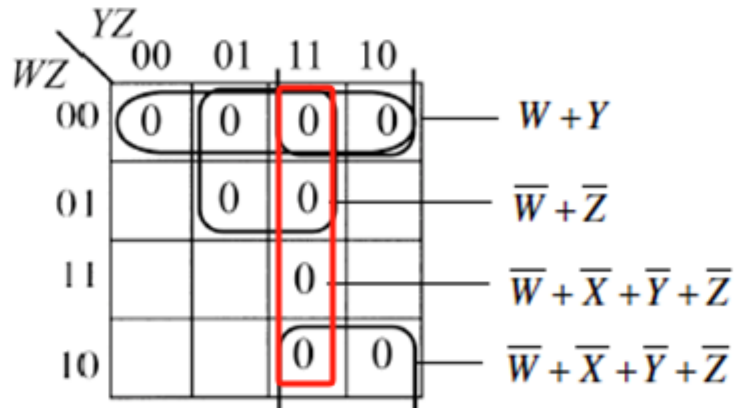
No reduction

(b) $(X+\overline{Y})(W+\overline{Z})(\overline{X}+\overline{Y}+\overline{Z})(W+X+Y+Z)$

$$= (W+X+\overline{Y}+Z)(\overline{W}+X+\overline{Y}+\overline{Z})(W+X+\overline{Y}+\overline{Z})(\overline{W}+X+\overline{Y}+Z)$$

$$(W+\overline{X}+\overline{Y}+\overline{Z})(W+\overline{X}+Y+\overline{Z})\cancel{(W+X+\overline{Y}+\overline{Z})}(W+X+Y+\overline{Z})$$

$$(\overline{W}+\overline{X}+\overline{Y}+\overline{Z})\cancel{(W+\overline{X}+\overline{Y}+\overline{Z})}(W+X+Y+Z)$$



(W + X)(W + Z^)(X + Y^)(Y^ + Z^)

$\cancel{(W+X)(\overline{W}+\overline{Z})(X+\overline{Y})(\overline{W}+\overline{X}+\overline{Y}+\overline{Z})}$

FIGURE 4-23